



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Friday, 21 August 2026, 3:09 PM**State** Finished**Completed on** Friday, 21 August 2026, 3:24 PM**Time taken** 15 mins 15 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d",&n);
6      int i=1;
7      int s=1;
8      int count=3;
9      while(s<=n)
10     {
11         count++;
12         i++;
13         count++;
14         s+=i;
15         count++;
16     }
17     printf("%d",count);
18     return 0;
19 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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